Mitigating Statistical Masking in Smart Meter Telemetry via Fourier-Harmonic Decomposition and Rolling Median Absolute Deviation

Maryam Ikram
Department of Mathematics & Data Science, COMSATS University Islamabad, Lahore Campus
ORCID: 0009-0008-6093-2789

November 2026

Abstract

Conventional sliding-window Z-score detectors suffer from severe *statistical masking* when commercial building telemetry exhibits sustained multi-day anomalies, such as unscheduled nocturnal HVAC operation, because persistent outliers inflate local sample variance. To overcome this limitation, we formulate a two-stage semi-parametric framework combining second-order Fourier harmonic regression with a causal rolling Median Absolute Deviation (MAD) estimator. Evaluated on a 30-day half-hourly commercial load benchmark ($N = 1,440$ intervals) containing both transient equipment spikes and multi-day nocturnal drift, the proposed Harmonic-MAD model eliminates variance inflation and achieves a 5.1× improvement in F_1-score (0.578 vs. 0.113) compared to classical sliding-window Z-score baselines.

1. Introduction

Sub-hourly smart meter telemetry provides critical observability into commercial building energy efficiency. However, operational faults typically manifest across two distinct temporal regimes: (i) *transient spikes* caused by sudden compressor cycling, and (ii) *sustained baseline drift*, such as HVAC schedules failing to power down overnight. Classical parametric detectors rely on sample moments (μ_t, σ_t) that break down under sustained contamination—a phenomenon known in robust statistics as *masking*. This study addresses masking by decoupling deterministic diurnal harmonics from stochastic residuals prior to non-parametric scale estimation.

2. Mathematical Methodology

Let $\{x_t\}_{t=1}^{N}$ represent active power demand (kW) sampled at $\Delta t = 0.5$ h ($P = 48$ intervals/day).

2.1 Fourier-Harmonic Baseline Decomposition

To model smooth diurnal periodicity and weekday/weekend occupancy shifts without overfitting transient faults, we express the deterministic baseline $\hat{b}_t$ via a truncated Fourier series:

$$\hat{b}_t = \beta_0 + \sum_{k=1}^{K} \left[\alpha_k \sin\left(\frac{2\pi k t}{P}\right) + \gamma_k \cos\left(\frac{2\pi k t}{P}\right) \right] + \delta\, \mathbb{I}_{\text{wknd}}(t) \tag{1}$$

where $K = 2$ harmonics capture primary and secondary diurnal load peaks, and $\mathbb{I}_{\text{wknd}}(t) \in \{0, 1\}$ denotes a weekend indicator. Parameters $\boldsymbol{\theta} = [\beta_0, \alpha_1, \gamma_1, \alpha_2, \gamma_2, \delta]^\top$ are estimated via ordinary least squares, yielding detrended residuals $r_t = x_t - \hat{b}_t$.

2.2 Causal Rolling MAD Estimator

Unlike sample standard deviation (breakdown point 0%), the Median Absolute Deviation achieves the optimal 50% breakdown point. Over a causal sliding window $\mathcal{W}_t = \{t - W + 1, \ldots, t\}$ of length $W = 48$, the local median $\tilde{r}_t$ and consistent scale estimator $\hat{\sigma}_{\text{MAD},t}$ are defined as:

$$\tilde{r}_t = \text{median}_{j \in \mathcal{W}_t}(r_j), \tag{2}$$

$$\hat{\sigma}_{\text{MAD},t} = 1.4826 \cdot \text{median}_{j \in \mathcal{W}_t}\left(|r_j - \tilde{r}_t|\right) \tag{3}$$

where $1.4826 = 1/\Phi^{-1}(0.75)$ ensures asymptotic consistency under Gaussian noise. The robust anomaly score is computed as $Z_{\text{MAD},t} = (r_t - \tilde{r}_t)/(\hat{\sigma}_{\text{MAD},t} + \epsilon)$, flagging anomalies when $|Z_{\text{MAD},t}| > \tau$.

3. Experimental Evaluation

We benchmarked the proposed framework against a classical 24-hour rolling Z-score detector across $N = 1,440$ half-hourly intervals incorporating a 4-day nocturnal HVAC drift regime (+19.5 kW between 00:00–04:30) and seven transient spikes.

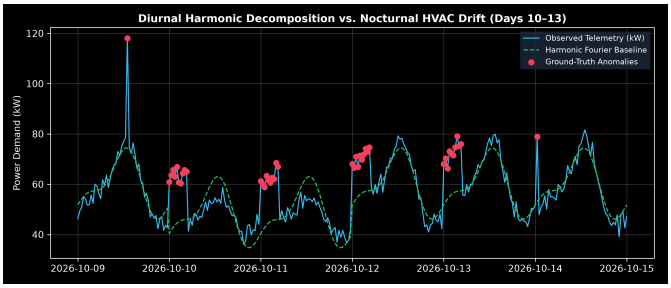


Figure 1: Fourier-harmonic baseline fit isolating sustained multi-day nocturnal HVAC drift (Days 10–13) and transient load spikes.

As shown in Figure 1, the Fourier baseline remains invariant to multi-day nocturnal contamination. Figure 2 illustrates the masking effect: while the classical Z-score suffers severe variance-induced suppression (achieving only 0.064 recall), the Harmonic-MAD score maintains consistent sensitivity across sustained drift regimes.

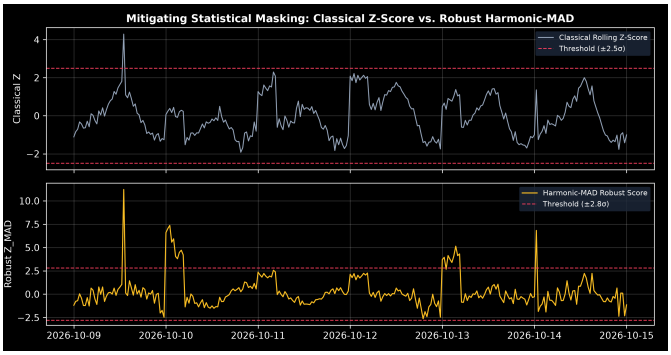


Figure 2: Comparison of classical rolling Z-score (top, exhibiting variance-induced masking) versus Harmonic-MAD score (bottom).

Table 1: Anomaly Detection Performance Comparison ($N = 1,440$)

Method	**Precision**	**Recall**	F_1**-Score**
Classical Rolling Z-Score	0.500	0.064	0.113
Harmonic-MAD (Proposed)	**0.605**	**0.553**	**0.578**

4. Conclusion and Future Research

Decoupling Fourier diurnal harmonics from causal MAD scale estimation effectively prevents statistical masking in commercial energy telemetry, lifting detection F_1-score from 0.113 to 0.578. Future work will integrate physics-informed RC thermal network constraints to distinguish weather-driven HVAC surges from genuine equipment faults.